Lab Exercise - 20 App Deployment Using ECS And ECR

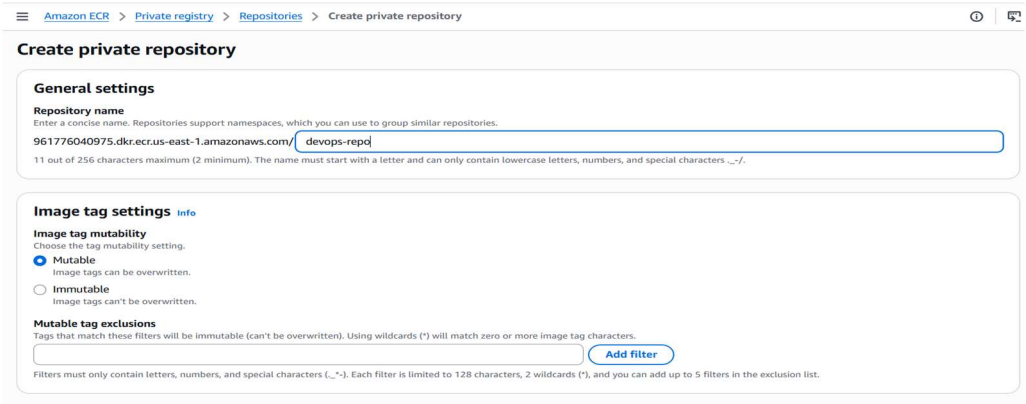
Name: Jiya Tyagi

SAP ID: 500119743

Batch: 2 DevOps

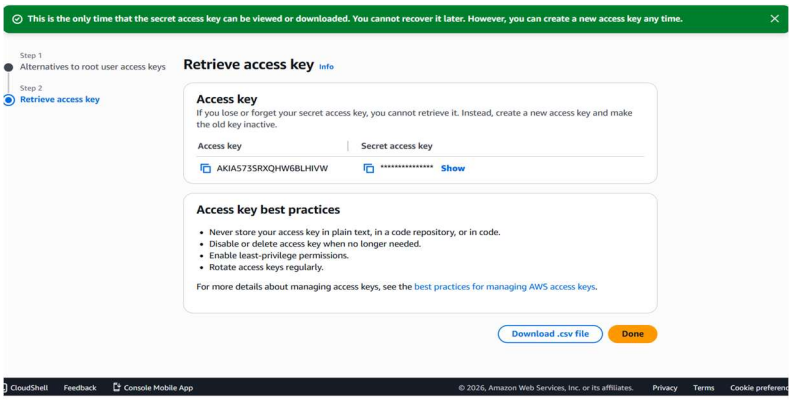
Step 1: Create ECR Repository

- Go to AWS Console
- Search Elastic Container Registry (ECR)
- Click Create Repository
- Give name (example: my-web-app)
- Click Create



Step 2: Configure aws

Create access key



Step 3: Authenticate Docker to ECR

Run this :

```
aws ecr get-login-password --region us-east-1 | docker login --username AWS --

password-stdin 961776040975.dkr.ecr.us-east-1.amazonaws.com
```

```
.dkr.ecr.us-east-1.amazonaws.com
Login Succeeded
```

Step 4: Build Docker Image

Run this :

```
docker build -t devops-repo .
```

```
[+] Building 25.1s (8/8) FINISHED                                                docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                0.1s
 => => transferring dockerfile: 102B                                                                0.0s
 => [internal] load metadata for docker.io/library/httpd:latest                                     4.3s
 => [internal] load .dockerignore                                                                   0.0s
 => => transferring context: 2B                                                                     0.0s
 => [1/3] FROM docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a505  18.1s
 => => resolve docker.io/library/httpd:latest@sha256:bdba5c86022f2d6ad0783168b07fe80541fe45e97b434b13ba8c959a5050  0.0s
 => => sha256:4e514480304900b8604d7662b05848e89e3563fe3b9551597ec54d4d48a4893f 289B / 289B                  0.8s
 => => sha256:61e1798e23ba12f14505e225391338db290ee6cec4fb6321c6c6b92ac8b546dcb 13.45MB / 13.45MB          7.2s
 => => sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf 2.00MB / 2.00MB             5.8s
 => => sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b60c110ee65c9c523b49c7 146B / 146B                 1.6s
 => => sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976 29.78MB / 29.78MB          16.2s
 => => extracting sha256:5435b2dcdf5cb7faa0d5b1d4d54be2c72a776fab9a605336f5067d6e9ecb5976                 0.6s
 => => extracting sha256:8261ede62f8c1c85c3e4b2ae3a9ae5a586b84fcc61b60c110ee65c9c523b49c7                 0.0s
 => => extracting sha256:4f4fb700ef54461cfa02571ae0db9a0dc1e0cdb5577484a6d75e68dc38e8acc1                 0.0s
 => => extracting sha256:a838a4be55ced5f2d7208c56a3a13ecbd15110574ee338337c50737aef163fdf                 0.1s
 => => extracting sha256:61e1798e23ba12f14505e225391338db290ee6cec4fb6321c6c6b92ac8b546dcb                 0.2s
 => => extracting sha256:4e514480304900b8604d7662b05848e89e3563fe3b9551597ec54d4d48a4893f                 0.0s
 => [internal] load build context                                                                       0.1s
 => => transferring context: 180B                                                                       0.0s
 => [2/3] WORKDIR /usr/local/apache2/htdocs/                                                            0.1s
 => [3/3] COPY index.html .                                                                             0.0s
 => exporting to image                                                                                  0.3s
 => => exporting layers                                                                                 0.1s
 => => exporting manifest sha256:84390a7b0d5796bd106e923e37c4c1daaa20de61b5536f9f885839e17110ec699       0.0s
 => => exporting config sha256:cc78baf051feefbfa212321bc55f3f27883c712b6fce14d43f589dc8a66b7240         0.0s
 => => exporting attestation manifest sha256:f01af328c1787db559a2e2193e92747631d8ab72f88f52fc8f0f619e9f99836d  0.0s
 => => exporting manifest list sha256:c2f3f6ce676e1506b20190ad0433ebb20b70ec904d9434541d0e4ce9df4e2578  0.0s
 => => naming to docker.io/library/devops-repo:latest                                                   0.0s
 => => unpacking to docker.io/library/devops-repo:latest                                                0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/hrjy6h8l81ii9tl7lohmwbumi
```

Step 5: Tag Docker Image

Run this :

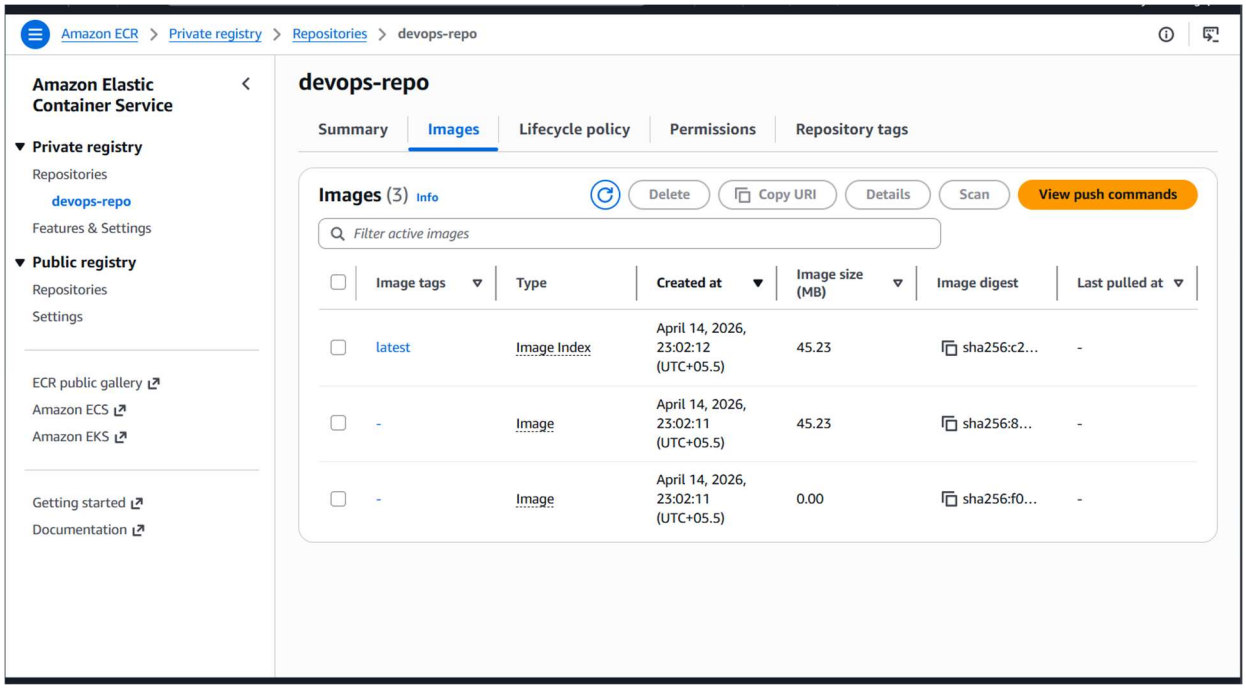
```
docker tag devops-repo:latest 961776040975.dkr.ecr.us-east-

1.amazonaws.com/devops-repo:latest
```

Step 6 : Push Docker Image

Run this :

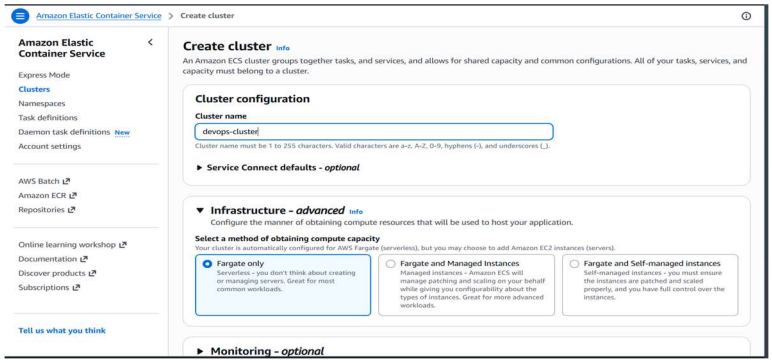
```
docker push 961776040975.dkr.ecr.us-east-1.amazonaws.com/devops-repo:latest
```

```
The push refers to repository [961776040975.dkr.ecr.us-east-1.amazonaws.com/devops-repo]
4e51448083049: Pushed
61e1798e23ba: Pushed
a77ba7d35f39: Pushed
a4ed997a6241: Pushed
4f4fb700ef54: Pushed
5435b2dcdf5c: Pushed
8261ede62f8c: Pushed
a838a4be55ce: Pushed
latest: digest: sha256:c2f3f6ce676e1506b20190ad0433ebb20b70ec904d9434541d0e4ce9df4e2578 size: 856
```



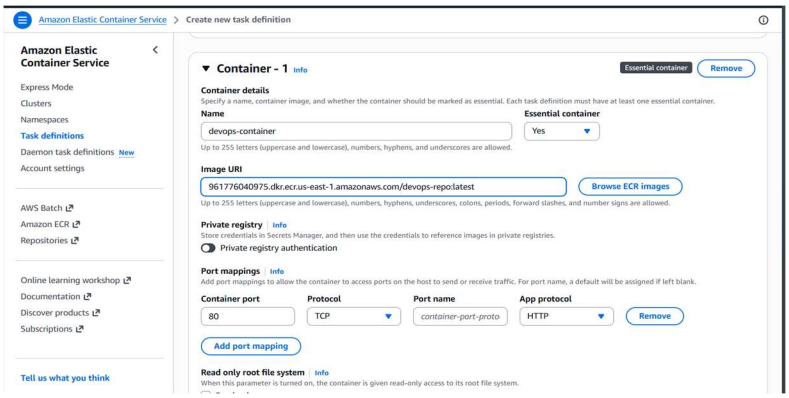
Step 7: Create Cluster

1. Go to ECS
2. Click Create Cluster
3. Select Fargate (Serverless)
4. Cluster Name: devops-cluster-new
5. Click Create



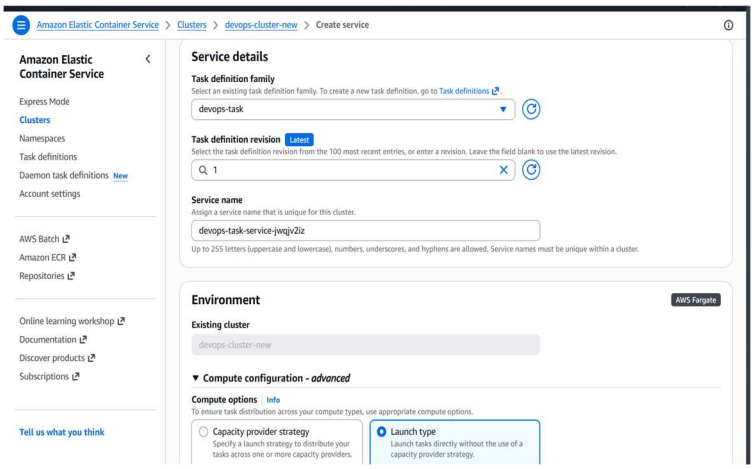
Step 8: Create Task Definition

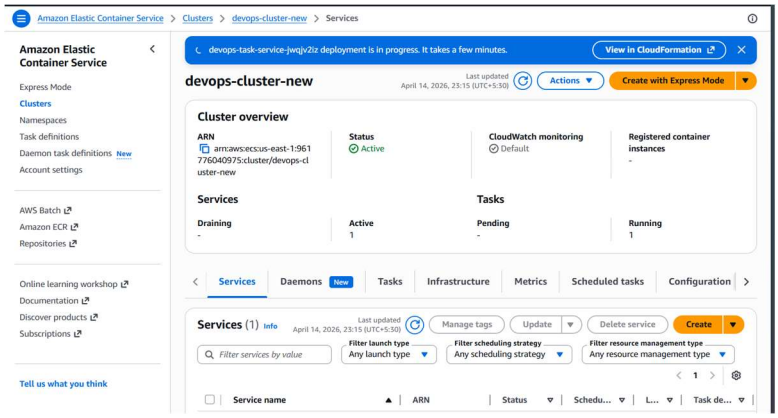
1. Launch Type → Fargate
2. Container Image → Paste ECR URI

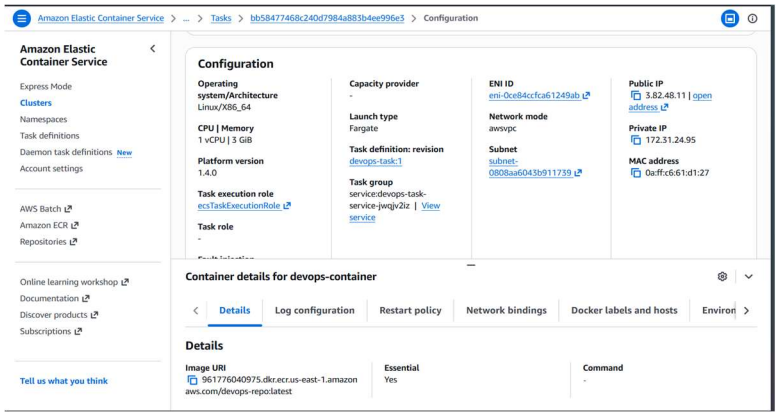


Step 9: Create Service

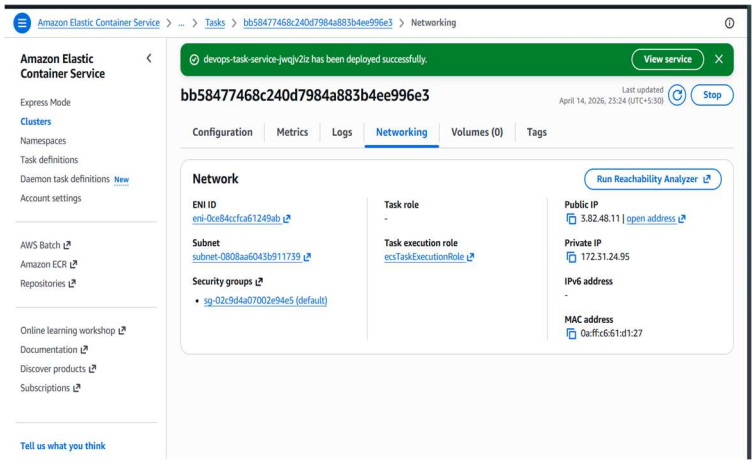
1. Go to Cluster
2. Click Create Service

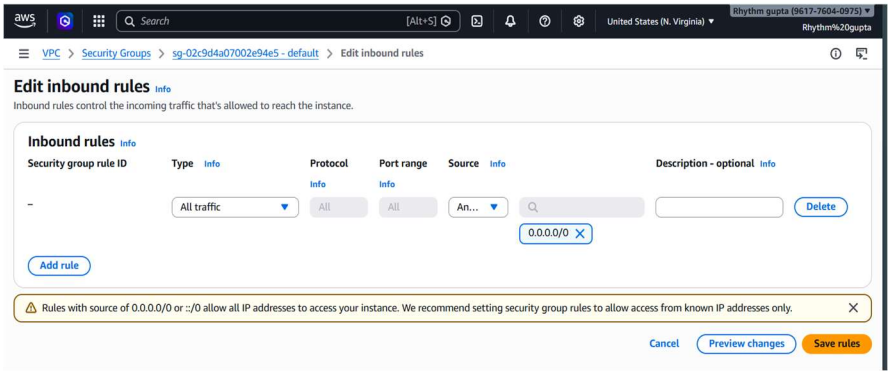


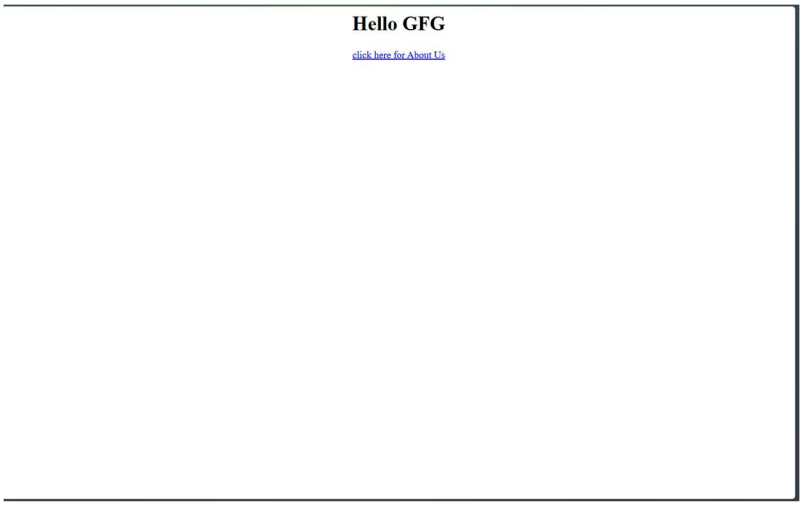




Step 10: Turn on all traffic in security broke





Step 11 : Verify Deployment

1. Go to Load Balancer
2. Copy DNS Name
3. Paste in Browser